Medium Access Control

- Gaining access to a LAN is a fundamental requirement for data transmission between stations:
  - This referred to *Medium Access Control* (MAC)
- There are two generalised strategies for controlling access:
  - *Centrally*: Stations must obtain permission before accessing the transmission medium
  - *Distributed:* Access is controlled by stations acting *together*

MAC Strategies

- *Centrally Controlled Access*:
  - Advantages:
    - The access logic required at each station is *simple.*
    - There is <u>no</u> co-ordination required between stations.
  - Disadvantages:
    - Single point of failure. If the controlling device develops a fault all stations are affected.
    - Potential *bottleneck.* All “requests for access” must be sanctioned by the central device.
- *Distributed Access* techniques are more commonly used and will be explored later

MAC Techniques

- There are <u>three</u> generalised access control techniques:
  - Reservation (based on a *Centralised* strategy - not covered)
    - Medium is divided into *time slots* which can be reserved by transmitting (Tx'ing) stations
    - Suited to stream traffic i.e. long and continuous transmissions on an *irregular* basis
  - Round Robin (based on a *Distributed* strategy):
    - Tx'ing stations take turns to transmit data
    - Efficient only if stations have a <u>lot</u> of data to transmit on a <u>regular</u> basis
    - Refer to slides on Token Ring
  - Contention (based on a *Distributed* strategy - see next slide)

MAC – Contention technique

- Characteristics of a *Contention* MAC technique:
  - Tx'ing stations *contend* for access when needed
  - The technique is *distributed* by nature
  - Works on a "*first come/first served"* basis and is easy to implement.
  - Very suited to *bursty* traffic i.e. short, sporadic transmissions.

Example technology using Contention - *Ethernet*

- *Ethernet* is a predominant *Bus/Star LAN* technology:
  - Employs a *Contention* MAC technique known as:

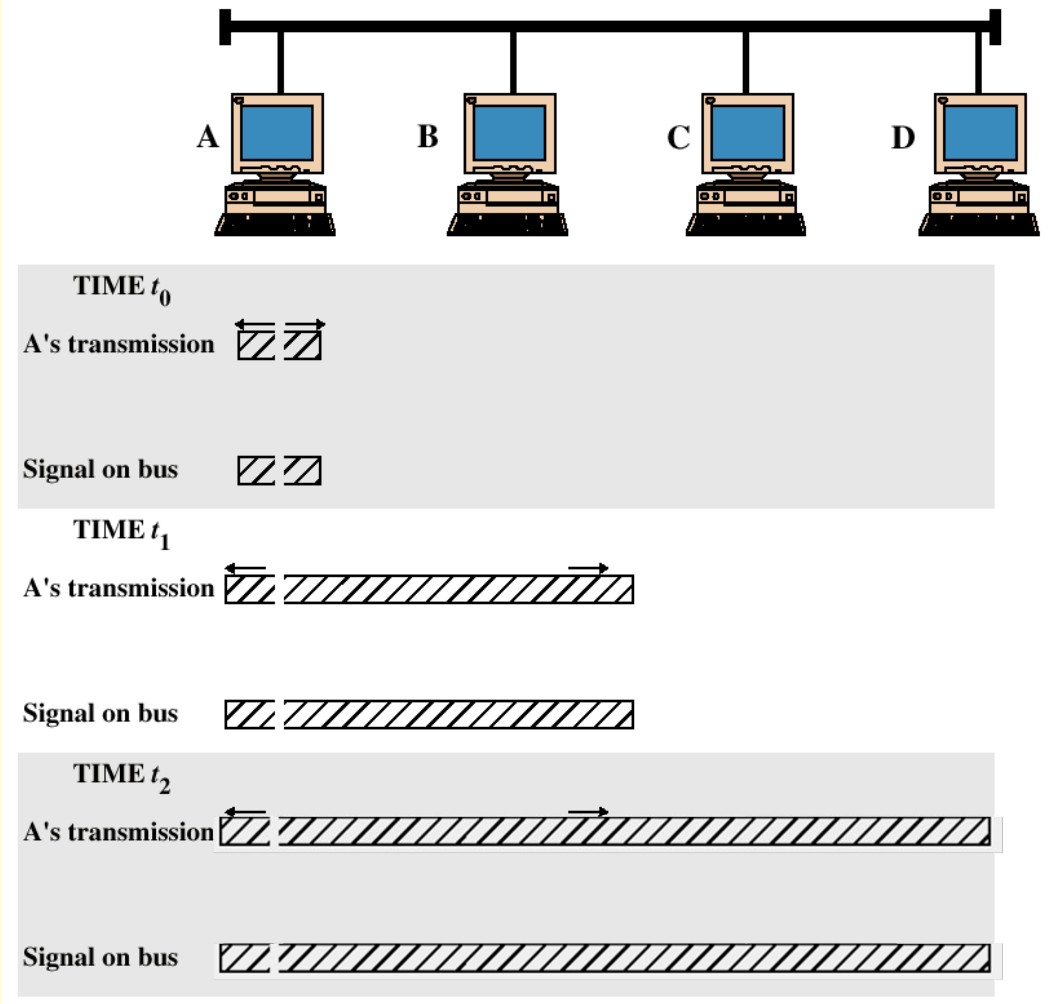
    Carrier Sense Multiple Access

    with ***Collision Detection (CSMA/CD)***
  - It is *random* access technique requiring Tx'ing stations to *contend* for access to the shared transmission medium
  - Most commonly used MAC technique on *bus* <u>and</u> *star* LANs.
  - Defined under IEEE802.3 standard

CSMA – Basic Operation

- To transmit a frame the Tx'ing station must:
  - Listen to the medium for any electrical activity - known as *carrier sensing*
  - If the medium is clear of frames, the station begins transmission
  - If the medium is in use, the station waits a random period of time before attempting to transmit the frame – known as a random *back-off period*

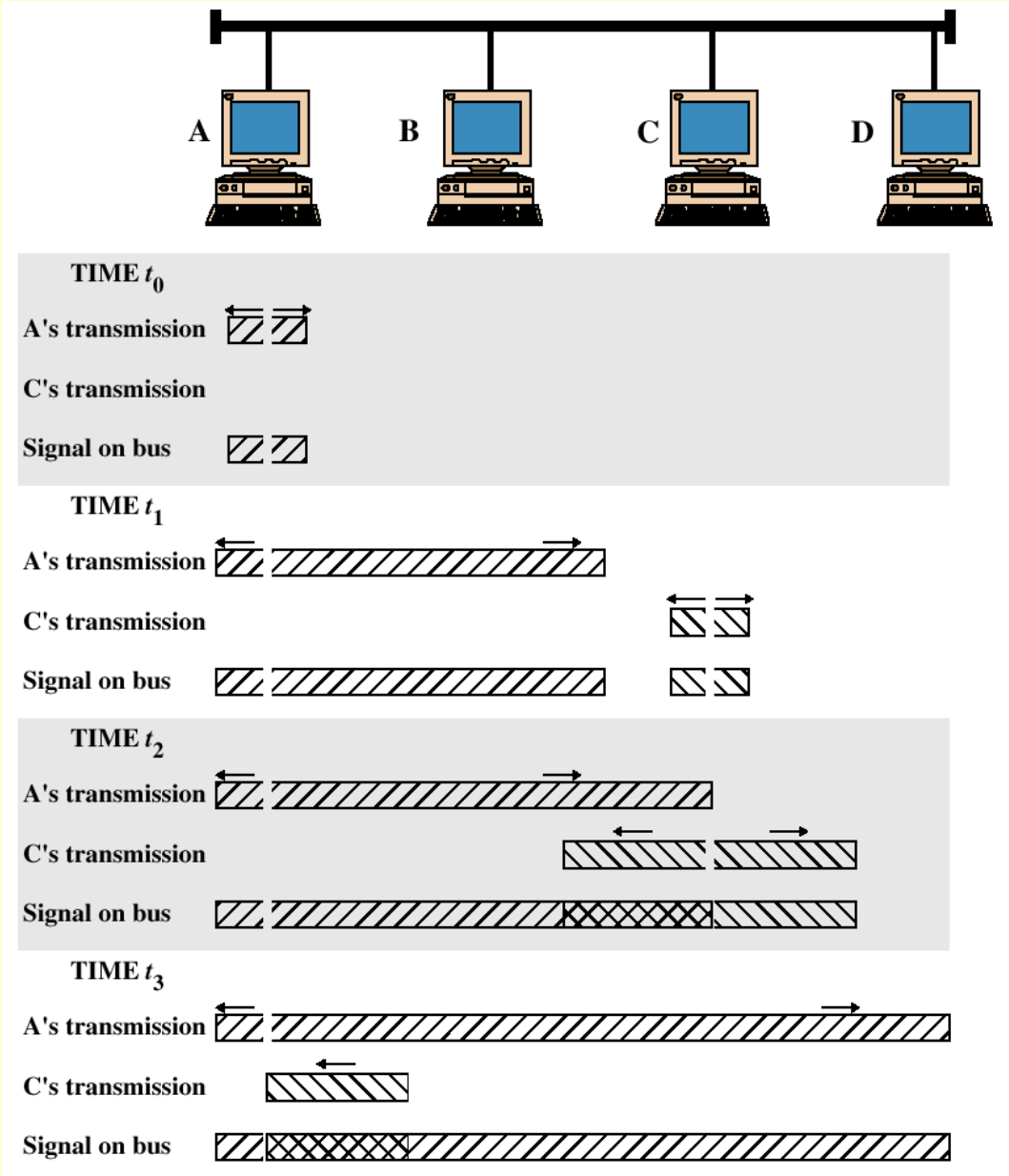
CSMA Normal Operation



CSMA – Basic Operation (contd.)

- In normal operation, Tx'ing stations behave like HDLC stations:
  - They send frames and occasionally expect ACKs
  - Recall ARQ techniques: *Go-Back-N* and *Selective Reject*
- There is a problem with this access technique:
  - Two or more stations may attempt to transmit at approx. the same time:
  - This will lead to a *collision* i.e. a *garbled* transmission.
    - Recall what happens when *digital* signals meet? They interfere with each other and become corrupted to the extend they are unreadable

CSMA Operation



CSMA – Basic Operation (contd.)

- Collisions are an example of *lost* data frames or *lost* ACKs:
  - Recall recovery scenarios from discussions on *Go-Back-N* and *Selective Reject*:
    - Frames involved in collisions will be detected either by:
      - The Receiver (Rx'er): If another frame is transmitted <u>after</u> the lost frame, this is detected as an *out-of-sequence* frames and a (S)REJ message is returned or,
      - The Tx'er: Recall a timer is set for each transmitted frame – ACKs must be returned before the timer expires otherwise the Tx'er is forced to send a *command* message (recall RR message with *Poll* bit set)

CSMA Characteristics

- CSMA on its own results in poor *network performance:*
  - *Utilization* of the link is low as only **one** data frame or ACK frame can exist on the medium – example of *half-duplex* mode
  - Collisions can occur leaving the medium unusable for the time taken to clear collisions
  - Tx'ing station relies on negative ACKs and/or expired timers to detect and fix problems
- To counteract these inefficiencies CSMA was extended to include *collision detection* (CSMA/CD)

CSMA/CD – Basic Operation

- As before, the Tx'ing station performs carrier sensing i.e. listening for electrical activity on the medium **before** attempting to transmit
- During transmission, the station continues to *listen* to the medium to detect any **collisions** - known as *Collision Detection*:
  - If detected:
    - The station *stops* transmitting the frame
    - It transmits a brief *jamming signal* to inform all other stations of the collision
    - It waits a random amount of time (*delta*) before attempting to retransmit the frame
    - Additonal collisions are dealt with using a *binary exponential backoff* (see explanation in class)

CSMA Versus CSMA/CD

- Without the use of Collision Detection, collisions would have to be dealt with using some form of *Error Control* technique:
  - The negative consequences of relying on *Error Control* techniques include:
    - The overall time to send multiple frames would be longer due to the use of *timers* and the re-transmission of *out-of-sequence* frames
    - Extra traffic would be generated on the network viz. ACKs and the re-transmission of *out-of-sequence* frames

CSMA Versus CSMA/CD - Contd.

- The use of *Collision Detection* has significant benefits:
  - The Tx'ing station resolves problems presented by collisions entirely on its own. The Tx'er only has to wait for collisions to disappear from the segment to fix the problems of *lost* data frames and lost ACKs rather than relying on *Error Control* techniques
  - All frames are delivered to the Rx'er, even after multiple attempts
  - *Error Control* can effectively be disabled as the Tx'ing station can assume frames have arrived at the Rx'er if no collisions occur
- In summary:
  - The overall time to send multiple frames is reduced significantly
  - *Link Utilization* is improved as network traffic is restricted to the transmission of data frames only

Collision Detection and *Frame Size*

- *Frame size* (length) is critical to the successful operation of *Collision Detection*:
  - Short frames prevent *collision* detection
  - Frames have to be a minimum length (see explanation and calculations in class).
  - Consequently, the MAC frame format includes a special field called *Pad.*
    - Extra octets ("random data") can be added to the frame.

Octets: 7	1	6	6	46 - 1500		4
Preamble	SOF	Destination MAC address	Source MAC address	LLC PDU (Data)	PAD	CRC

802.3 MAC Frame

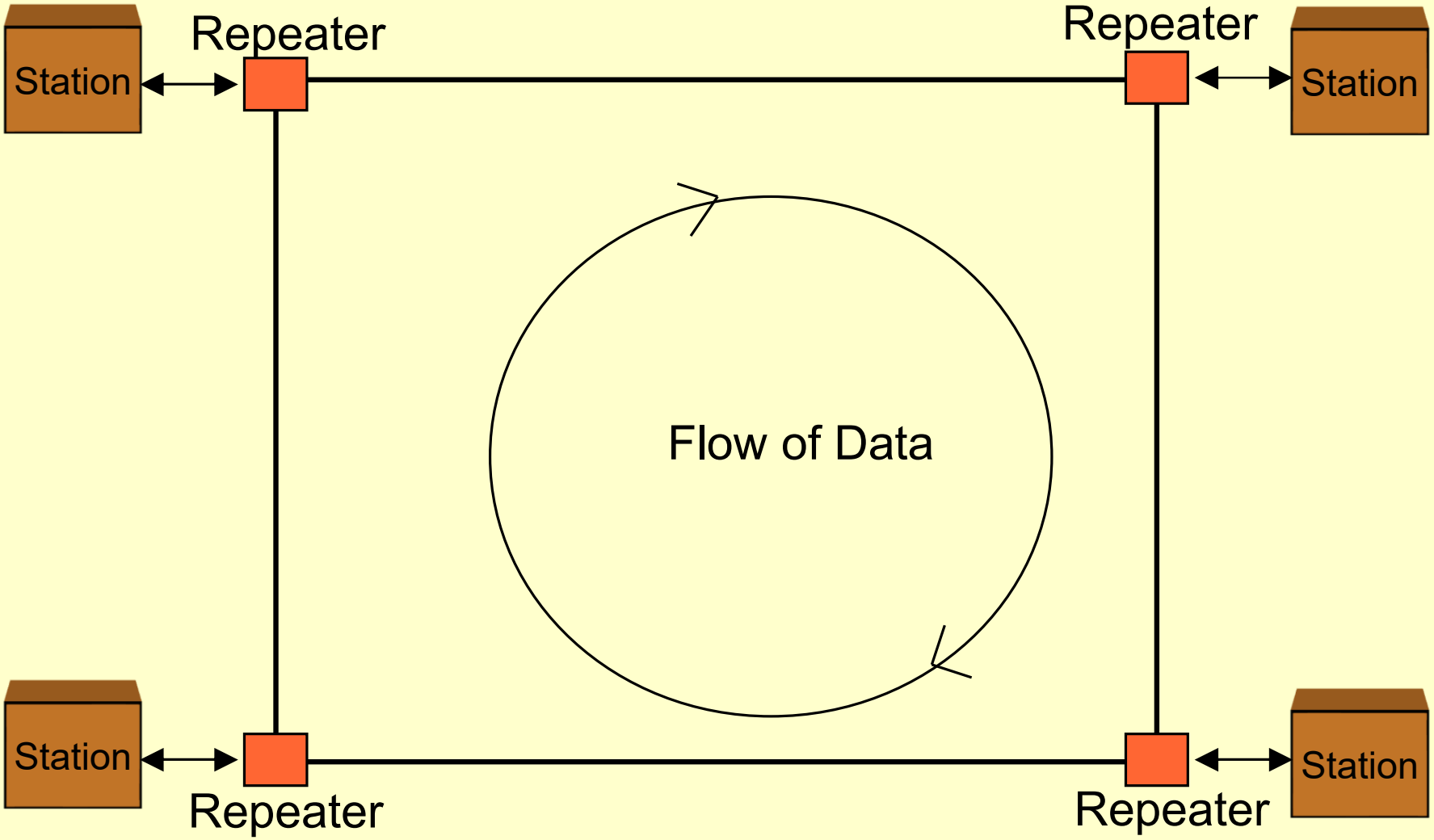
CSMA/CD Simple Algorithm (Not Examinable)

```
WHILE (Do always - Infinite loop)
    IF Frame on Tx medium;
        Check Destination Address field;
        IF (own address)
            Copy frame;
        ELSE ignore frame;
    WHILE (Have Frame to Transmit)
            Listen to Transmission medium;
            IF Clear;
                Commence frame Transmission;
                WHILE (Frame is transmitting)
                    Continue to Listen to Transmission
                    medium;
                    IF Collision detected;
                            Stop transmission;
                            Transmit Jamming signal;
                            Wait random period to
                                Retransmit;
            ELSE wait random period to transmit;
    END Transmit WHILE Loop;
END Infinite WHILE Loop;
```

Ring LANs

- Consists of a number of stations each of which is connected to a *repeater.*
- These in turn are connected to two other repeaters by *unidirectional* transmission links to form a ring.
- Data are transferred *bit-by-bit* around the ring from one repeater to the next.
- The repeaters *regenerate* and *retransmit* <u>each</u> bit.

Ring LANs


Station
Repeater
Repeater
Station
Flow of Data
Station
Repeater
Repeater
Station

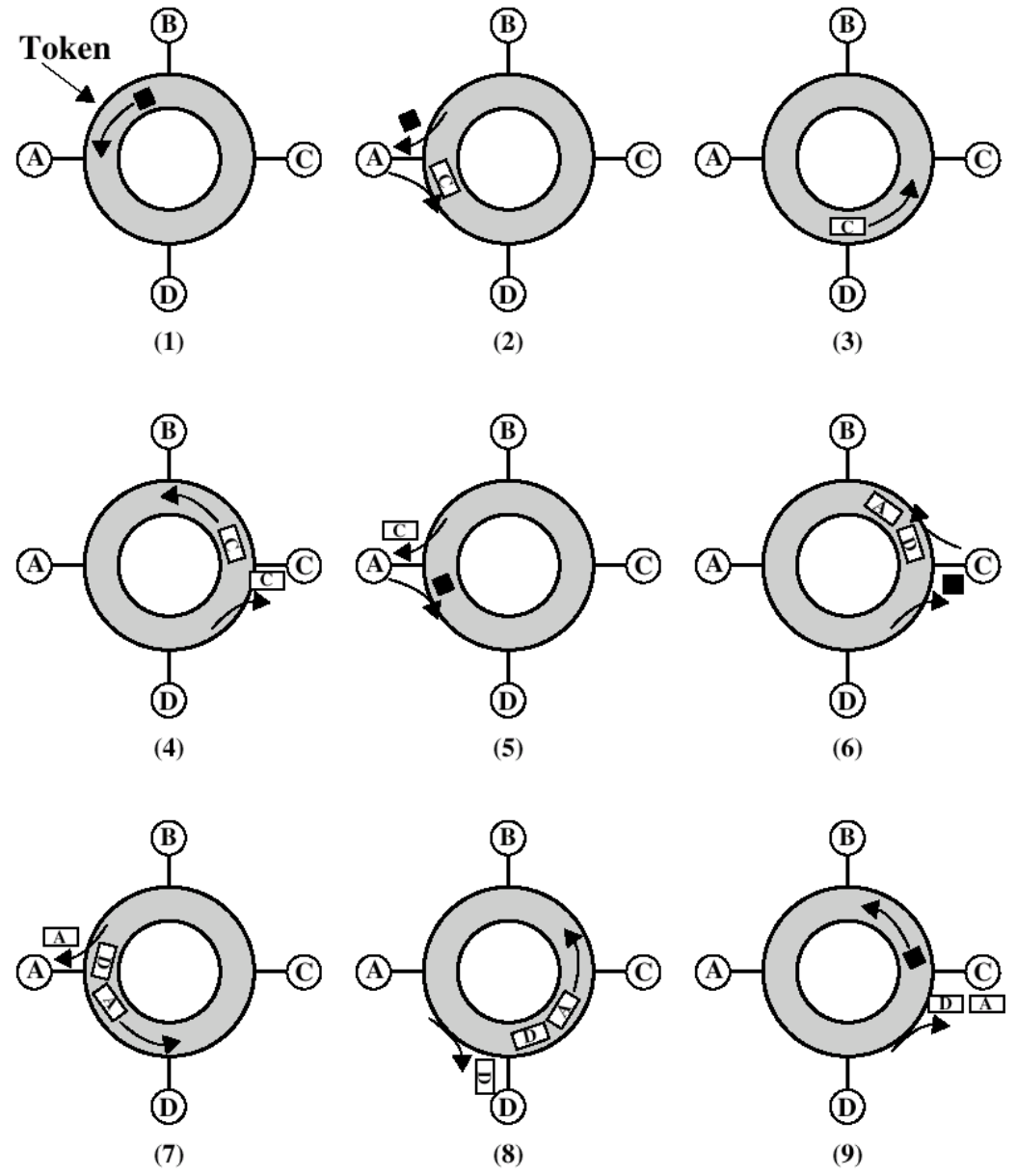
Ring LANs – Repeater Functionality

- Repeaters perform <u>three</u> functions:
  - *Frame insertion*: *Frames* (with addresses) are placed onto the Tx medium by the repeater.
  - *Frame reception*: As a frame passes through a repeater the address field is copied to the station. If the station recognises the address, the entire frame is copied.
  - *Frame removal*: A repeater can remove a frame by not repeating it to the next link:
    - Either the *addressed repeater* can remove the data or, the *transmitting station* can (after one round trip).

Example Access Technique - *Token Ring*

- The most commonly used MAC technique for *ring* LANs is *IBM Token Ring:*
  - This is a *Round Robin* technique and is defined under the IEEE802.5 standard.
- Token Ring uses a small frame, called a *Token:*
  - The token circulates around the *ring* indefinitely
  - A station wishing to transmit must <u>seize</u> the token and remove it from the LAN

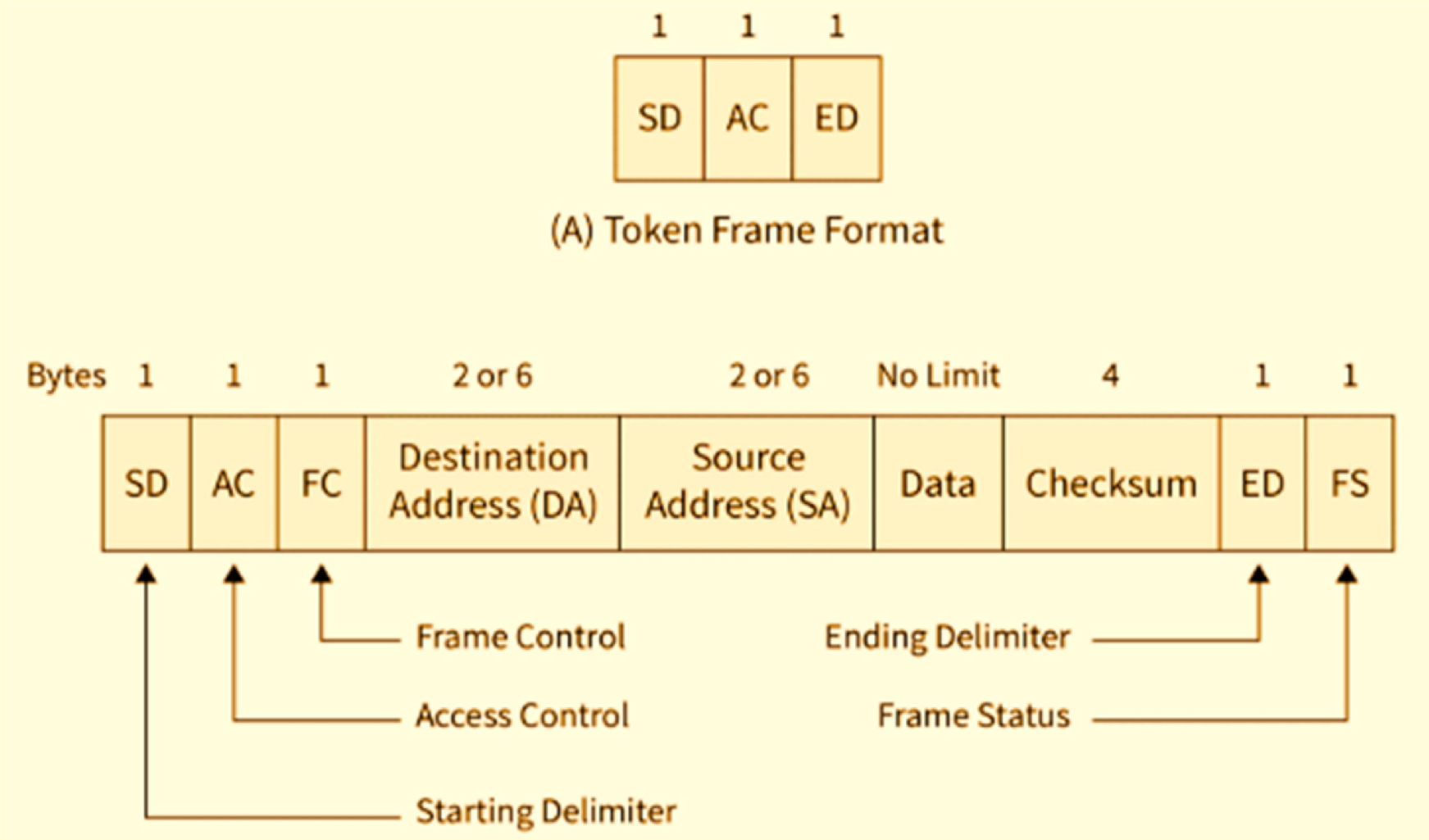
Token Ring Operation



Token Ring – Basic Operation

- The *token* is then transformed into a *start-of-frame* field for a *data frame* (a single bit is changed – see next slide):
  - More fields are added to construct a *data frame*
  - The token is effectively removed from the ring and replaced by a ***Data*** *frame*
  - When the frame returns to the *transmitting* station it removes the *frame* from the ring and inserts a new token

802.5 MAC Frame Format



Token Ring - Simple Algorithm (Not Examinable)

WHILE (Do always - Infinite loop)
- IF Frame on Tx medium;
  - Check Destination Address field;
  - IF (own address)
    - Copy frame;
  - ELSE ignore frame;
- WHILE (Have Frame to Transmit)
  - IF Token available;
    - Remove Token;
    - Transmit frame;
    - While (Frame in Transit);
      - If Frame returns;
      - Remove Frame;
      - Place Token on Tx'ion medium
    - End "Frame Transit" WHILE Loop;
  - End "Have Frame to Transmit" WHILE Loop;

END Infinite WHILE Loop;

Token Ring Advantages/Disadvantages

- Advantages:
  - Access is *efficient* and *fair* under heavy traffic loads.
- Disadvantages:
  - Access can be inefficient under light traffic loads:
    - A station must wait for a token before transmitting.
  - There is a requirement for *token maintenance*:
    - Tokens can be *lost* or *duplicated.*
    - One station is assigned responsibility for token maintenance.

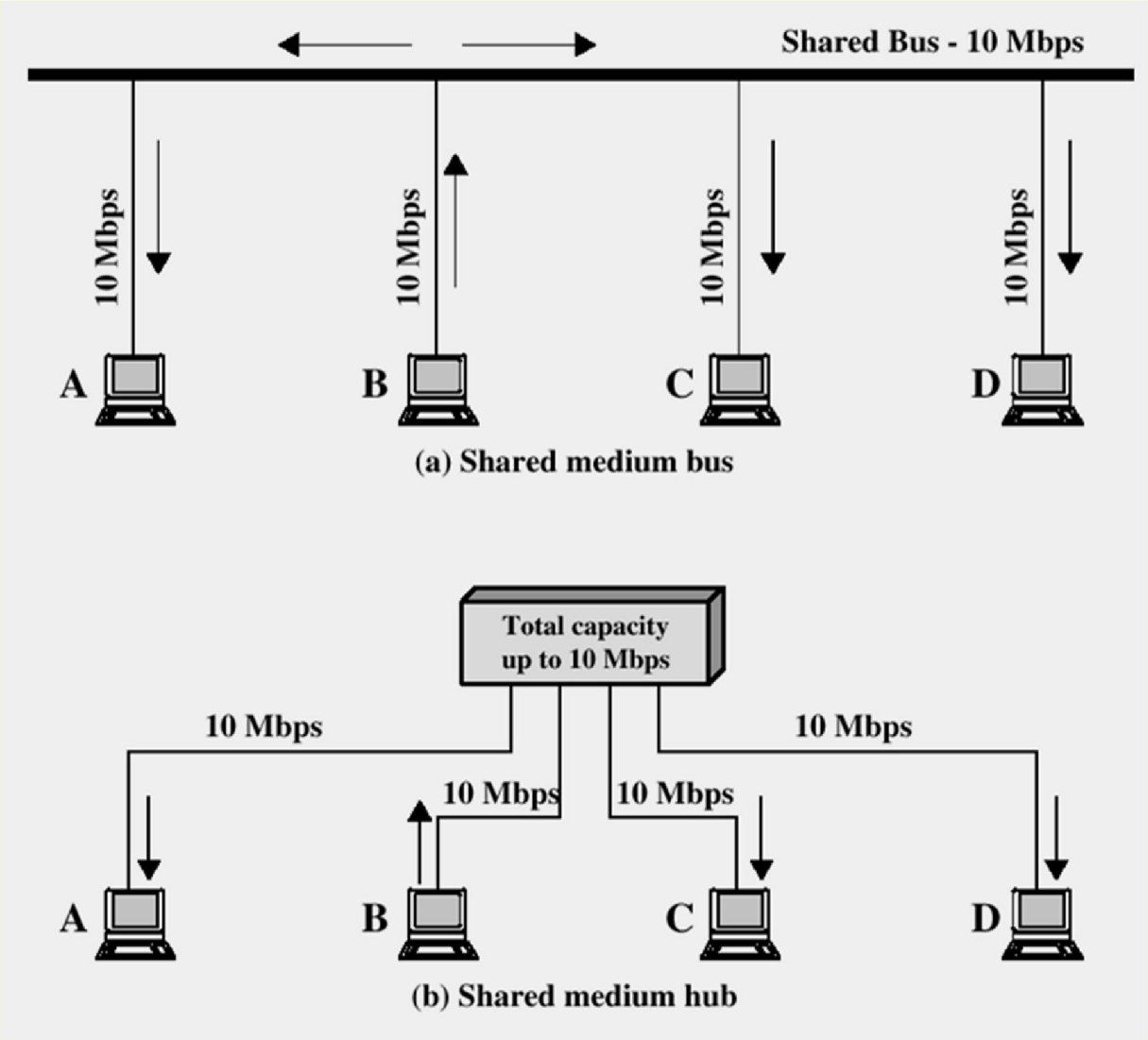
Star LANs

- Uses a central component known as a *hub*
- Two types of hub:
  - *Shared-medium hub* and,
  - *Switched LAN hub*

Shared-medium hub LAN

- *Shared-medium hub (Hub-star LAN)*
  - Transmission from any station is repeated to <u>all</u> stations
  - Only <u>one</u> station can transmit at any one time
  - This is the same behaviour as a Bus Lan
    - Hence this type of *star* LAN is also known as a *star-shaped bus* (as <u>opposed</u> to a *simple bus* LAN)
  - Advantages of *star-shaped bus* LAN:
    - Can use existing building wiring

Hub-Star LAN Configuration



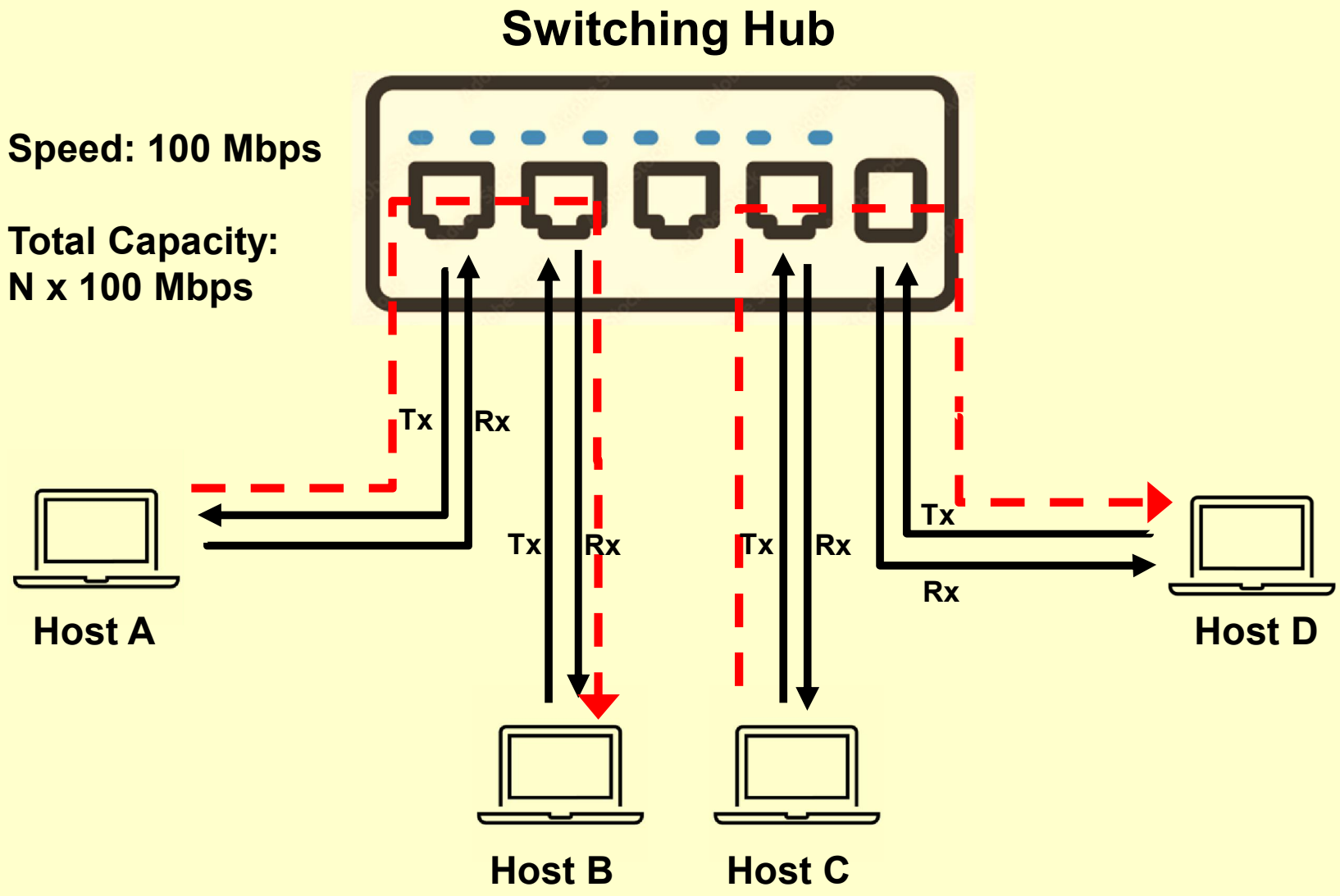
Switched LAN hub LAN

- *Switched LAN hub (aka Switched LA or Switched-star LAN)*
  - Acts as a switch i.e. incoming frame is only passed to the addressed station
  - Frames from <u>other</u> stations can be switched simultaneously
- <u>Greater</u> performance can be achieved with a *switched LAN hub* <u>over</u> a *shared-medium hub*

Switching LAN Operation

- The operation of a *Switched LAN* hub device differs from a *Shared-medium* hub device
- On a *Switched* LAN:
  - Each host connects to the switch at a ***switch port*** using <u>separate</u> transmit and receive links (Tx & Rx)
  - The switch knows which host is connected to which port (see Hosts A, B, C & D on previous slide)
  - The ports employ *store-and-forward* technology

Switched LAN Configuration



Switching LAN Benefits

- Using separate Tx & Rx links facilitates *full-duplex* communications mode:
  - Hosts can send and receive data at the same time
  - Bus LANs & Shared medium LANs only allow for *half-duplex* mode as all hosts share the same medium

Switching LAN Benefits

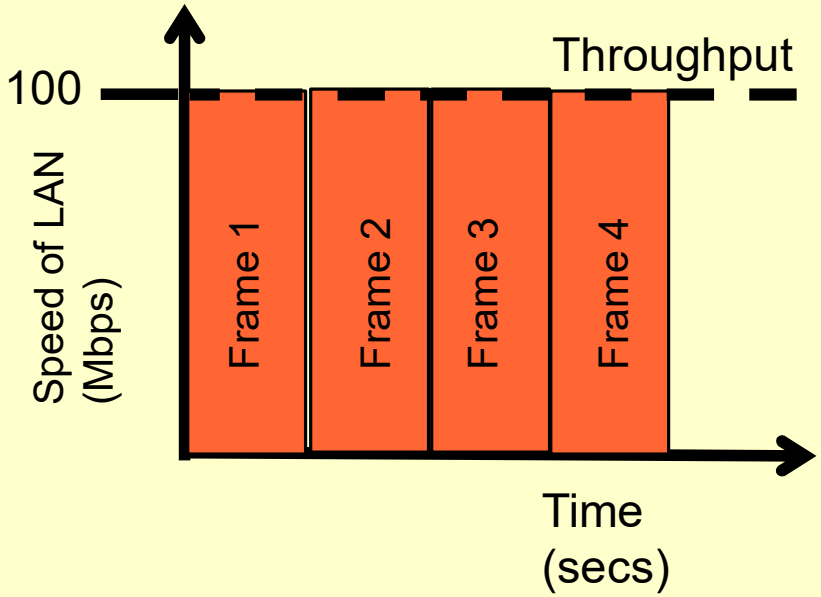
- Each host connected to a dedicated port allows for <u>multiple</u> *simultaneous* communications to pass through the switch:
  - In the diagram above, Hosts A & B can exchange frames at the same time as Hosts C & D <u>without</u> collisions – *contention free*
  - This is a significant performance improvement over Bus LANs and Shared medium LANs which only allow for a <u>single</u> frame transmission with the possibility of collisions - *contention*

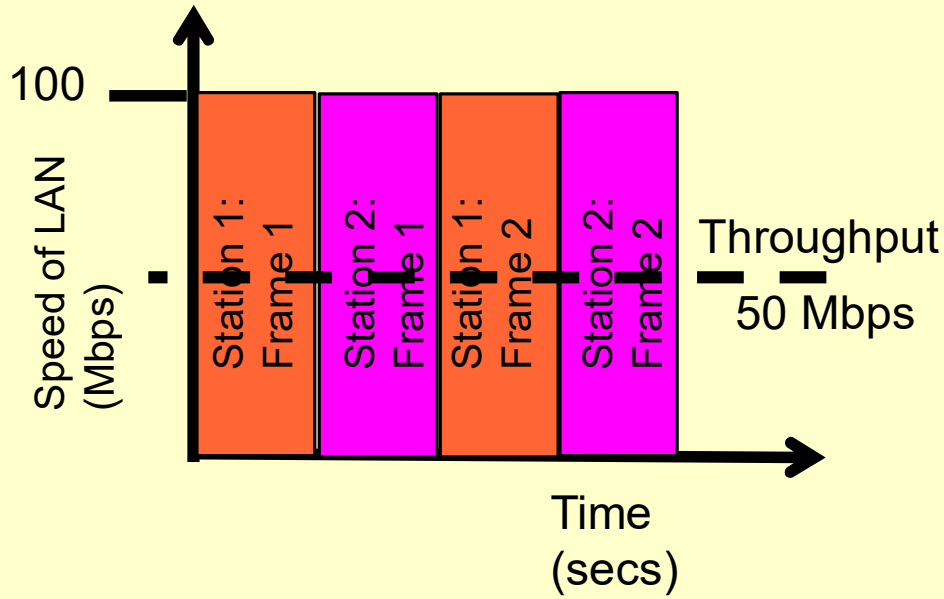
Switching LAN Benefits

- The use of *store-and-forward* technology allows for:
  - The switch to transfer frames to the correct port. Frames are stored on an incoming port and the switch decides which outgoing port the frame is to be moved to
  - Frames originating on different hosts can be sent to the same destination host without collision. The frames are simply stored sequentially in the destination host's outgoing port and are sent when the link is clear

Star LANs – Performance consideration

- **Contention-free** Switched-LANs provide improved performance:
  - As simultaneous communications between pairs of stations can be facilitated





Single station transmitting <u>without</u> contention on a Switched LAN experiences matching *throughput* and *speed* (100 Mbps)

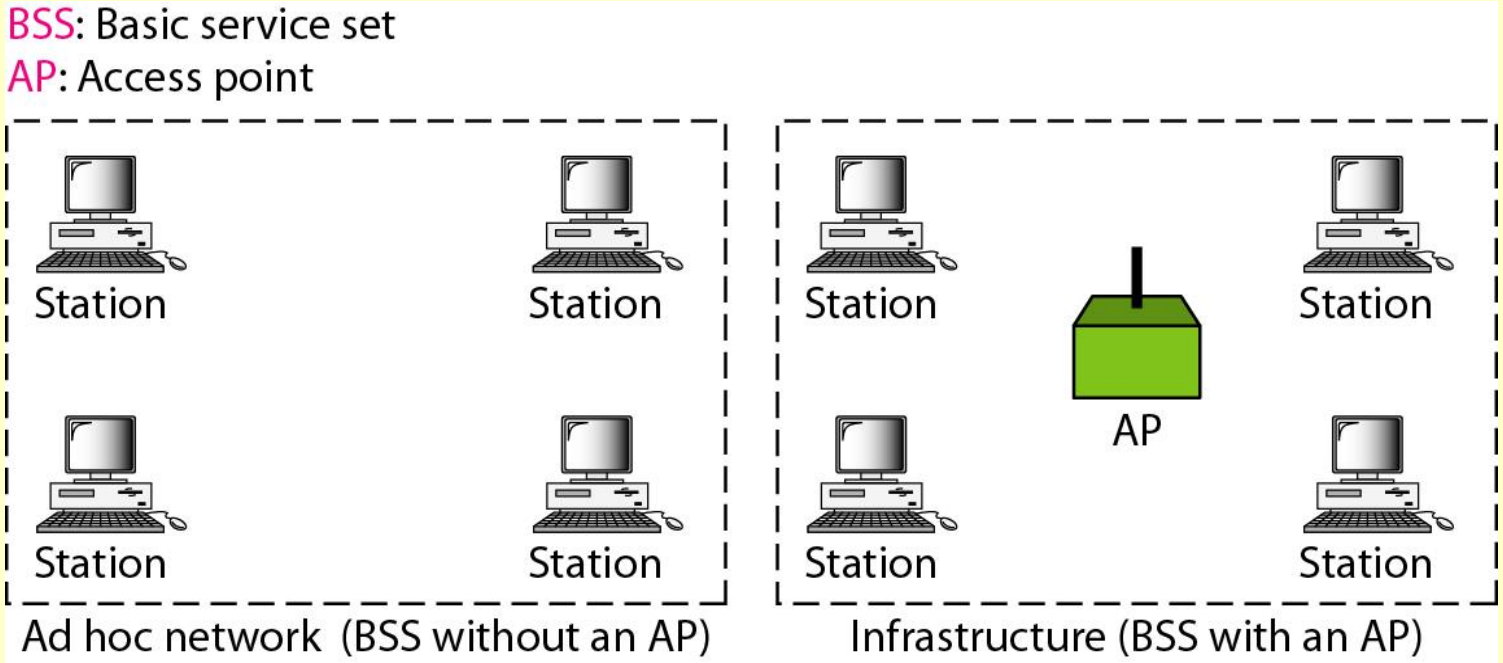
Two stations transmitting <u>with</u> contention on a Bus or Shared-medium LAN experiences reduced *throughput* of approx. 50 Mbps.

Wireless LAN

- The specification for wireless LANs is addressed under the *IEEE 802.11* sub-committee.
- This specification defines two types of architecture:
  - The Basic Service Set (BSS), and,
  - The Extended Service Set (ESS).

BSS Wireless LAN

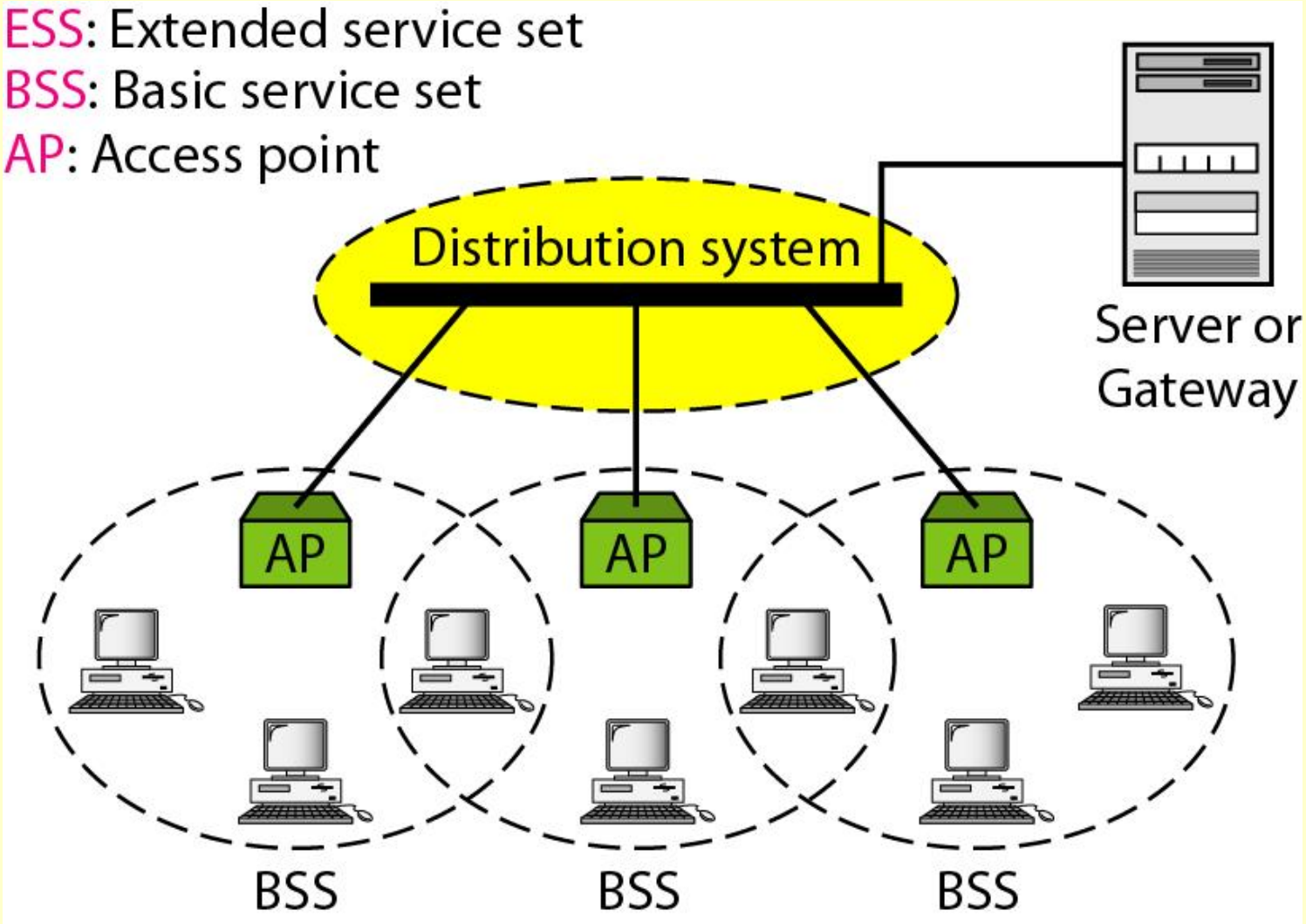
- The BSS typically comprises either:
  - A number of mobile stations that work independently
  - A number of mobile stations inter-working with a central **base** station known as an *Access Point* (AP)



ESS Wireless LAN

- An Extended Service Set (ESS) typically comprises two or more Infrastructure BSSs (with APs).
- The APs are usually connected via a *distribution network* (typically a wired LAN but can be wireless).
- Like BSSs, ESSs define two types of stations:
  - Stationary stations which usually connect via the wired LAN. These stations include the APs.
  - Mobile stations. Some of these stations can only communicate within a single BSS, others between BSSs and yet others between ESSs.

ESS Wireless LAN



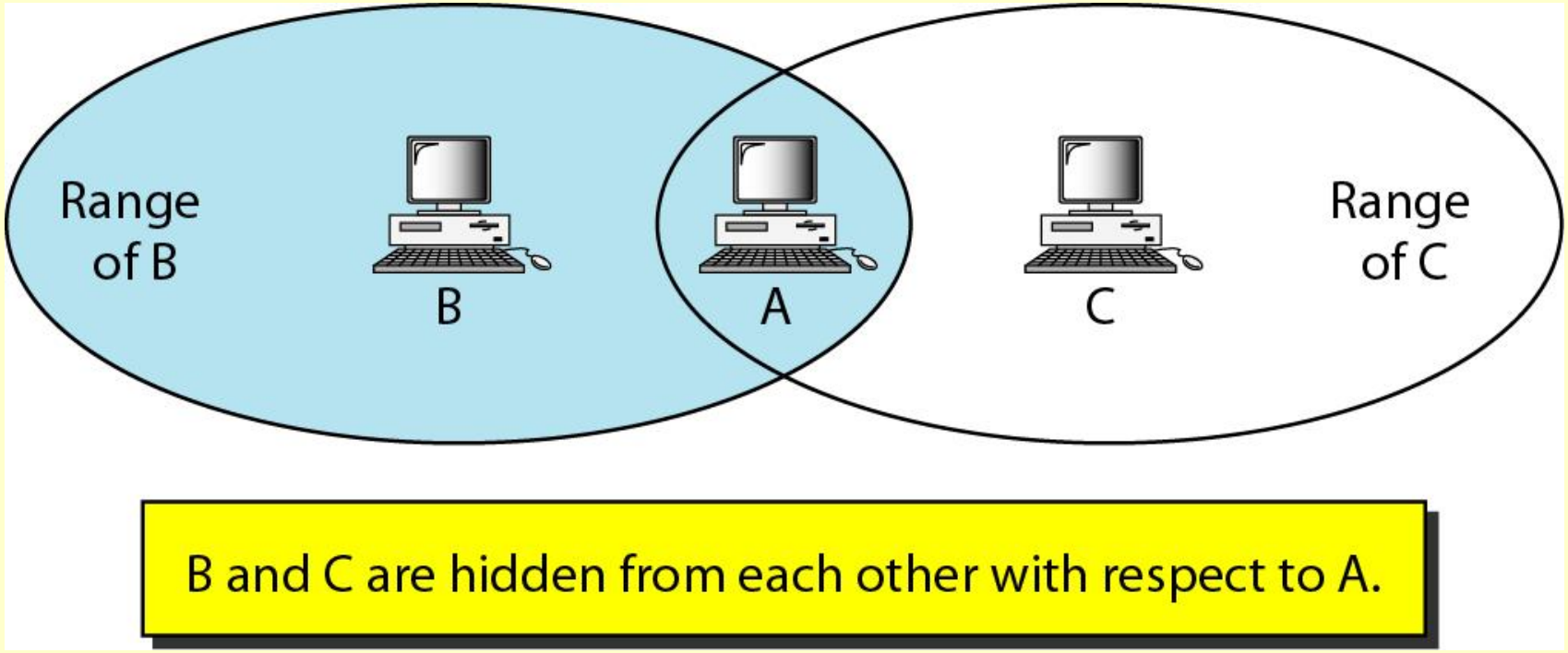
Wireless LAN Access Technique

- The access technique employed on a wireless LAN is ***CSMA/CA*** (Collision Avoidance)
- CSMA is similar to that used on Wired LAN:
  - Before a Tx'ing station transmits a frame, it must *listen* to the shared medium i.e. the air interface. In this case, it's listening for *radio signal* activity
  - As with wired LANs:
    - Timers and Back-off periods are used in the event that the medium is in use
    - The use of CSMA means that collisions can and do occur

Collisions on a WLAN

- Collision Detection (CD) on a **Wired** LAN relies on a Tx'ing station being able to detect a collision
- This is not always possible on a **Wireless** LAN:
  - The distances between stations may prevent <u>all</u> stations receiving <u>all</u> transmissions on the LAN as per the *Hidden Station Problem* (refer to explanation in class)
- Consequently, the strategy used to address collisions on a Wireless LAN is different:
  - It involves trying to avoid collisions in the first instance
  - The technique is known as *Collision Avoidance* and when used with CSMA it's called: ***CSMA/CA***

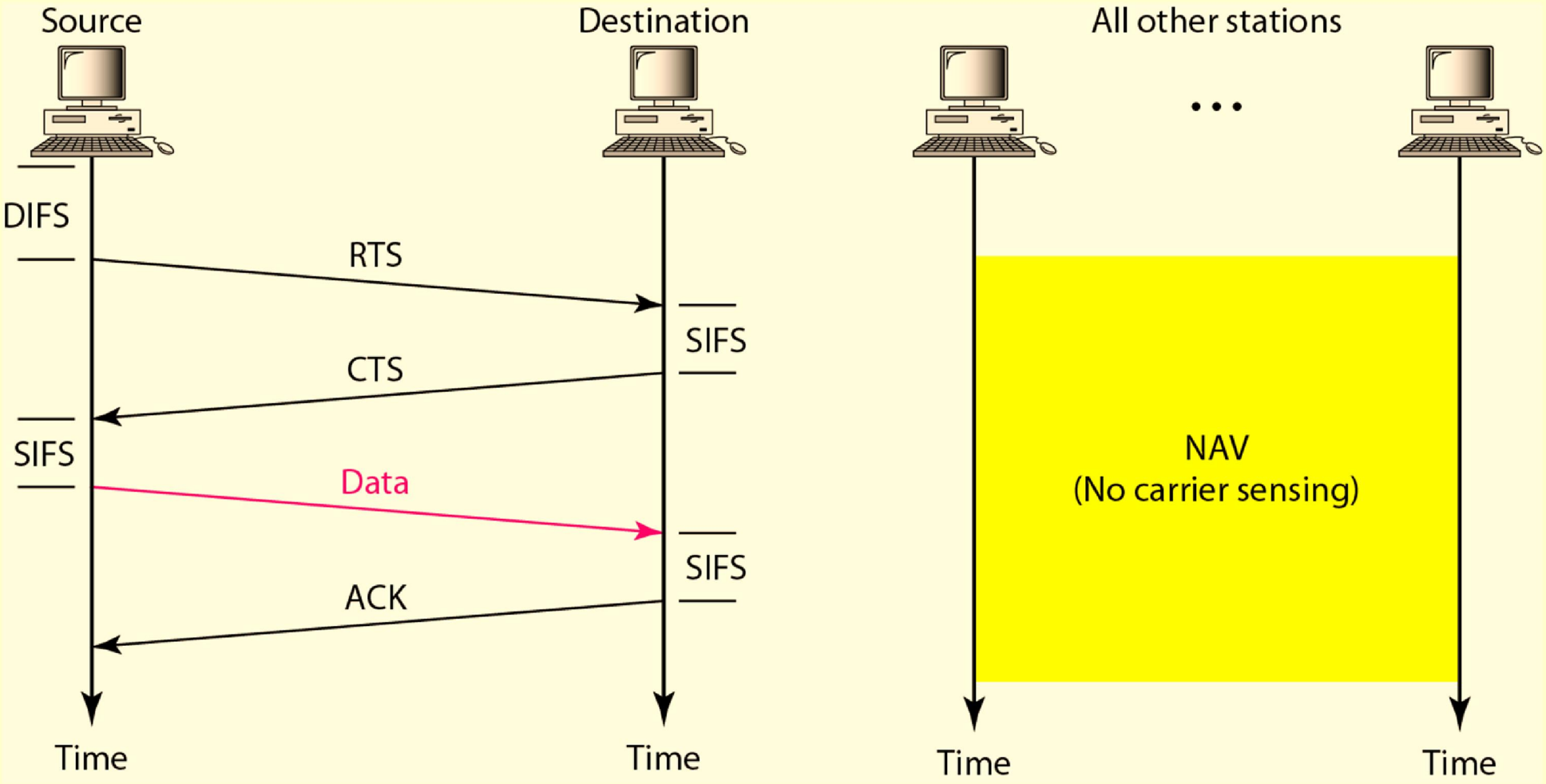
The *Hidden Station* Problem



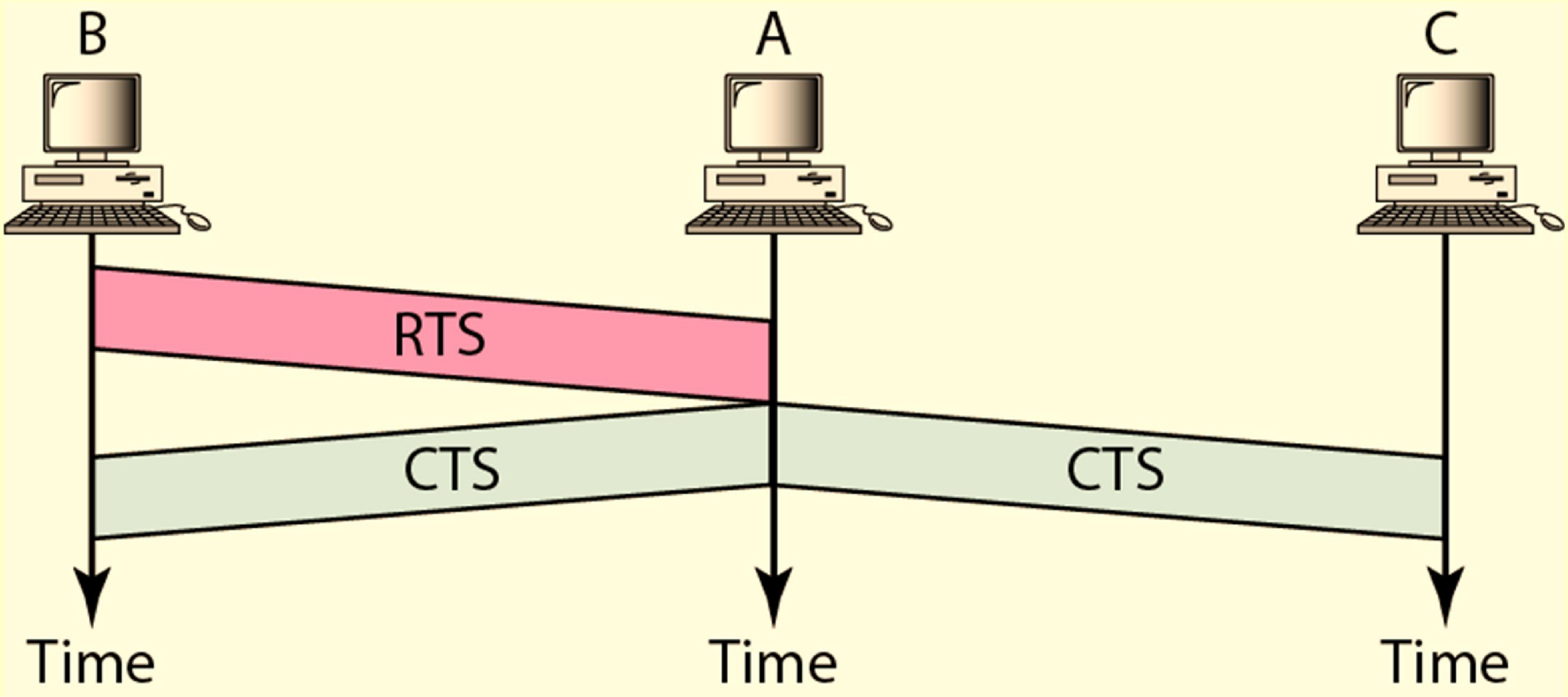
CSMA/CA Basic Operation

- CSMA/CA uses fixed *timers* and *control message exchange*
  - Message exchange technique uses two messages: *Ready To Send* (**RTS**) and *Clear To Send* (**CTS**)
  - The timers used are called: *Short Inter-Frame Space* (**SIFS**), *Distributed Inter-Frame Space* (**DIFS**) and, *Network Allocation Vector* (**NAV**)
  - These will be explained in class with reference to following slides

Operation of CSMA/CA



CSMA/CA for Hidden Station



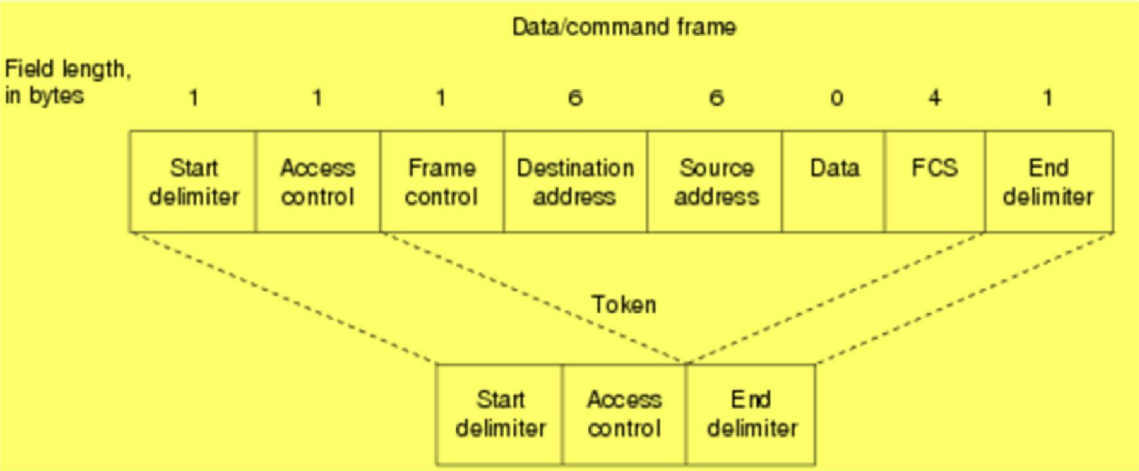
MAC Frame Formats

802.3 MAC Frame

Octets: 7	1	6	6	46 - 1500	4
Preamble	SOF	Destination MAC address	Source MAC address	LLC PDU (Data)	PAD	CRC

802.11 MAC Frame

Octets : 2 bytes	2 bytes	6 bytes	6 bytes	6 bytes	2 bytes	6 bytes	0 to 2312 bytes	4 bytes
FC	D	Address 1	Address 2	Address 3	SC	Address 4	Frame body	FCS

802.11 MAC Header

802.5 MAC Frame